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Physics-informed neural networks with orthogonal collocation for chemical kinetics simulation

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ABSTRACT

Accurate dynamic modeling is essential for chemical process optimization. While physics-informed neural networks (PINNs) offer a promising differential equation solver, conventional PINNs can be difficult to train for nonlinear and multiscale systems. To address this, we propose an Orthogonal Collocation Physics-Informed Neural Network (OCPINN), in which time-varying interpolation coefficients are predicted by a neural network and combined with Lagrange basis functions constructed at Gauss–Legendre interpolation nodes. OCPINN introduces a structured solution representation while retaining end-to-end automatic differentiation. The method is validated on an industrial methanol-to-propylene (MTP) fixed-bed reactor case and two benchmarks, including a nonlinear consecutive reaction and the stiff Robertson problem. Case studies demonstrate improved accuracy of the proposed OCPINN compared with conventional PINNs and variable-scaling PINNs. OCPINN achieves the lowest mean state-scaled global root mean square error in all three cases, with pronounced improvements for the nonlinear reaction and MTP reactor and a modest improvement for the Robertson problem. These results support OCPINN as a structured physics-informed representation for nonlinear and multiscale chemical-process models.

1. Introduction

The increasing use of model-based monitoring, optimization, and control in the chemical industry demands accurate and reliable dynamic modeling of reaction systems (Zhao et al., 2025; Yin et al., 2023). These complex chemical processes ranging from petrochemical refining to pharmaceutical synthesis, are governed by mass and energy conservation laws. They are typically modeled by sets of ordinary differential equations (ODEs) or differential-algebraic equations (DAEs), which can be high-dimensional, strongly nonlinear, and may even exhibit pronounced stiffness (Kim et al., 2021; Goswami et al., 2024). Mature numerical integration methods, including backward differentiation formulas, implicit Runge–Kutta schemes, and adaptive stiff solvers, provide accurate and reliable solutions for many of these systems and remain the preferred choice for one-off forward simulations. However, in process systems engineering applications such as parameter estimation, dynamic optimization, model predictive control, and repeated scenario evaluation, the dynamic model may need to be evaluated or differentiated repeatedly within an iterative computational workflow. These applications motivate the development of continuously differentiable and physics-consistent model representations that can complement, rather than universally replace, established numerical integrators.

Within this context, physics-constrained machine learning (PCML) has emerged as an important branch of scientific machine learning, in which physical knowledge is incorporated into the model architecture, training objective, or coupling between mechanistic and data-driven components. Mukherjee and Zavala (Mukherjee and Zavala, 2026) recently summarized PCML approaches in chemical engineering into broad categories including soft-constrained learning, hard-constrained learning, and hybrid or gray-box modeling. These formulations have been investigated for surrogate modeling, process monitoring, uncertainty quantification, optimization, and control (Mukherjee and Zavala, 2026).

Within this broader PCML landscape, standard physics-informed neural networks (PINNs) typically adopt a soft-constrained formulation for solving forward and inverse differential-equation problems. By incorporating governing-equation residuals together with initial and/or boundary conditions into the training objective, PINNs construct continuous neural representations that approximately satisfy the

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prescribed physical model (Raissi et al., 2019). PINN-based methods have consequently been investigated in process systems engineering for state estimation (Yan et al., 2025), chemical reactor modeling (Wu et al., 2024), process-related transport phenomena (Horiguchi et al., 2025), optimal operation (Wang and Wu, 2024), reaction kinetics (Zhang et al., 2024), polymerization systems (Ryu et al., 2024), and dynamic process modeling (Velioglu et al., 2025). For forward problems in which the governing equations are known, PINNs can be trained without interior reference-solution data, with the differential equations themselves providing the primary training signal. Their continuously differentiable representation is also attractive for applications requiring sensitivities or gradient-based calculations. The practical implementation of PINNs is limited by several numerical and optimization challenges. The simultaneous minimization of multiple residual components can generate poorly balanced optimization landscapes, particularly when state variables or governing equations span widely separated scales. These difficulties can become more pronounced in stiff or multiscale systems, where conventional PINNs may exhibit slow convergence, strong training oscillations, or large and conflicting residual gradients (Cuomo et al., 2022; Sharma et al., 2023). In addition, conventional PINNs generally impose physical relations through soft penalties and therefore do not guarantee exact satisfaction of equality or inequality constraints. Recent hard-constrained PCML methods have addressed this issue from a different direction. For example, Iftakher et al. (Iftakher et al., 2026) proposed KKT-Hardnet, in which nonlinear equality and inequality constraints are enforced through a differentiable projection layer derived from Karush–Kuhn–Tucker conditions.

A range of structural and algorithmic strategies has also been developed specifically to improve neural solution methods for transient, multiscale, and stiff dynamics. De Florio et al. (De Florio et al., 2022) strictly enforced initial constraints using the extreme theory of functional connections (X-TFC). Physics-informed random projection neural networks (PIRPNN) (Fabiani et al., 2023) and variable-scaling strategies such as (VS-PINN) (Ko and Park, 2025) have been proposed to simplify the learning task and mitigate scale imbalance and spectral-bias effects. Furthermore, integrating concepts from classical time-integration theory has been explored for stiffness-dominated dynamical systems. Architectures such as the deferred correction network (DCNet) (Jeon et al., 2024) and implicit single-step neural ODEs (Fronk and Petzold, 2024) have been shown to improve training behavior and convergence for stiff dynamical systems. For multi-scale applications, neural operator-based approaches (Goswami et al., 2024) learn solution operators without step-by-step time integration, enabling rapid evaluation of stiff chemical kinetics. Collectively, these studies demonstrate that physics-informed dynamic modeling can be improved through constraint treatment, variable scaling, architectural modifications, implicit integration, and operator-based representations, while also highlighting the value of incorporating established numerical-analysis principles into scientific machine-learning formulations.

Spectral and collocation techniques have likewise been introduced into neural differential-equation modeling. Earlier studies explored spectral discretization and collocation-based Neural ODE training as alternatives to repeated sequential integration (Quaglino et al., 2020; Roesch et al., 2021). More recently, Shapovalova and Tsay (Shapovalova and Tsay, 2025) formulated Neural ODE training as a fully discretized simultaneous optimization problem. In their formulation, a neural network represents the unknown right-hand side of the ODE, while collocation coefficients describing the state trajectory and the neural-network parameters are optimized simultaneously from observed trajectory data using a nonlinear programming solver. Lueg et al. (Lueg et al., 2026) extended the simultaneous approach to neural differential-algebraic systems, in which neural networks represent unknown relationships within an otherwise partially specified DAE model. Orthogonal collocation is used to fully discretize the dynamic constraints, and the state trajectories and neural-network parameters are determined through a simultaneous nonlinear-programming formulation that can additionally incorporate algebraic and inequality constraints. Importantly, these Neural ODE and Neural DAE studies address a different learning problem from that considered here: their neural networks are used to identify unknown dynamical or constitutive relationships from data, whereas the governing equations in the present work are assumed to be fully specified and the neural network is used to approximate their solution.

Orthogonal collocation nevertheless offers a useful numerical structure for constructing physics-informed solution representations. Classical orthogonal collocation approximates a differential-equation solution using interpolation polynomials constructed over carefully selected nodes, such as Gauss, Gauss–Radau, or Gauss–Lobatto points (Young, 2019; Carey and Finlayson, 1975; Finlayson, 1974). The resulting basis representation provides high-order approximation for sufficiently smooth solutions and converts continuous differential equations into structured algebraic relations (Çevik et al., 2025). In conventional orthogonal collocation and orthogonal collocation on finite elements, a finite set of state or polynomial coefficients is determined for the discretized trajectory. In the simultaneous Neural ODE and Neural DAE formulations described above, these coefficients or state values are optimization variables coupled directly with the neural-network parameters. In contrast, the present work addresses a forward initial-value problem with fully specified governing equations. The neural network maps the independent coordinate to time-varying coefficients associated with a Legendre–Lagrange basis, which are used to reconstruct the continuous state trajectory before evaluation of the governing-equation residual. Unlike the fixed coefficients defining a conventional collocation polynomial within each element, these coefficients vary with the evaluation coordinate, so the reconstructed solution is not restricted to a polynomial of degree N-1. Its derivatives include both coefficient variation and basis variation, retained through end-to-end automatic differentiation. Orthogonal collocation therefore serves here as a structured solution-reconstruction mechanism within residual-based physics-informed learning, rather than as a full discretization used to construct a simultaneous nonlinear-programming problem.

In this study, we propose an Orthogonal Collocation Physics-Informed Neural Network (OCPINN) for the solution of ODE systems. The principal methodological distinction is the introduction of a Legendre–Lagrange reconstruction layer in which the neural network predicts time-varying coefficient functions instead of directly outputting the state variables as in a conventional PINN. The reconstructed continuous solution is differentiated using automatic differentiation and trained using only the governing-equation residual and initial-condition loss; no interior reference-solution data are used for the forward problems considered in this study. We investigate whether this structured representation improves solution accuracy and optimization behavior relative to a conventional PINN for nonlinear reaction dynamics, the full stiff Robertson system without quasi-steady-state reduction, and an industrial methanol-to-propylene fixed-bed reactor model.

2. Methodology

This section introduces the proposed Orthogonal Collocation Physics-Informed Neural Network (OCPINN) method. The framework introduces a Legendre–Lagrange reconstruction structure inspired by orthogonal collocation into physics-informed neural-network solution learning. The following subsections describe the network architecture and the Legendre–Lagrange reconstruction formulation, followed by the construction of the loss function and the training procedure. The evaluation metrics used to quantify model accuracy are subsequently introduced.

2.1. Network architecture and numerical formulation

The overall structure of OCPINN consists of three components: a feedforward neural network module, a Lagrange interpolation–based reconstruction module, and a physics-constrained loss function, as illustrated by the blue, yellow and orange blocks in Fig. 1, respectively. Unlike a conventional PINN that directly predicts the state variables, the neural network in OCPINN predicts time-varying coefficient functions $\{c_i(\tau)\}_{i=0}^{N-1}$. The continuous solution is then reconstructed as a weighted combination of Lagrange basis functions defined on Gauss–Legendre interpolation nodes. The reconstructed state and its derivatives are subsequently used to evaluate the governing-equation residuals through automatic differentiation. This hybrid representation provides a structured basis representation of the solution and couples the predicted dynamics through the Legendre–Lagrange interpolation functions.

Consider a system of first-order ordinary differential equations:

$$\frac{dy(t)}{dt} = f(y(t), t), \quad t \in [t_0, t_f], \quad y(t_0) = y_0 \tag{1}$$

where $y(t)$ denotes the state variable to be solved, $f(y(t), t)$ is a given vector-valued function that describes the temporal evolution of the system, and $t \in [t_0, t_f]$ represents the time domain. The initial condition is specified as $y(t_0) = y_0$. Although the formulation is illustrated in the context of time-dependent ordinary differential equations, the proposed method can be readily extended to spatial domains.

The physical time domain in OCPINN is first normalized by mapping the interval $[t_0, t_f]$ to the standard domain $\tau \in [-1, 1]$, which is consistent with the definition domain of the Legendre polynomials. The mapping is given by

$$\tau(t) = \frac{2(t - t_0)}{t_f - t_0} - 1 \tag{2}$$

The Legendre polynomials are orthogonal over the interval $[-1,1]$, and their roots satisfy optimal numerical quadrature properties. They can be defined analytically as

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}\left[(x^2 - 1)^n\right], \quad n = 0, 1, 2, \ldots \tag{3}$$

Within this interval, N Legendre nodes $\{\tau_i\}_{i=0}^{N-1}$, corresponding to the roots of the Legendre polynomial $P_N(\tau)$, are selected as interpolation nodes for constructing the Lagrange basis. The solution $y(t)$ can then be expressed through Lagrange interpolation as

$$y(t) = \sum_{i=0}^{N-1} c_i(\tau(t)) \ell_i(\tau(t)) \tag{4}$$

where $c_i(\tau) \in R^{N_d}$ represents the interpolation coefficient associated with the i-th interpolation node. Unlike fixed values in classical polynomial interpolation, $c_i(\tau(t))$ are functions of the evaluation coordinate generated by the neural network. Their values are determined through physics-residual minimization and the initial-condition treatment. N_d denotes the dimensionality of the ODE state vector. $\ell_i(\tau)$ denotes the Lagrange basis function (Gao and Zahr, 2023), defined as

$$\ell_i(\tau) = \prod_{\substack{j=0 \\ j \neq i}}^{N-1} \frac{\tau - \tau_j}{\tau_i - \tau_j} \tag{5}$$

which satisfies $\ell_i(\tau_j) = \delta_{ji}$ (the Kronecker delta function), that is,

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases} \tag{6}$$

2.2. Loss function formulation and training protocol

The training objective of OCPINN/PINN is to minimize two types of errors over the entire temporal domain: the physics-based residual loss $\mathcal{L}_{\text{phys}}$ and the initial condition loss $\mathcal{L}_{\text{ic}}$.

$$\mathcal{L}_{\text{total}} = \lambda_{\text{phys}} \cdot \mathcal{L}_{\text{phys}} + \lambda_{\text{ic}} \cdot \mathcal{L}_{\text{ic}} \tag{7}$$

where the weighting coefficients λ_{phys} and λ_{ic} are set to 1. The physics loss is defined as according to the governing equations:

$$\mathcal{L}_{\text{phys}} = \sum_{d=1}^{N_d} \omega_d \mathcal{L}_d, \quad \mathcal{L}_d = \frac{1}{N_k} \sum_{k=1}^{N_k} r_{k,d}^2 \tag{8}$$

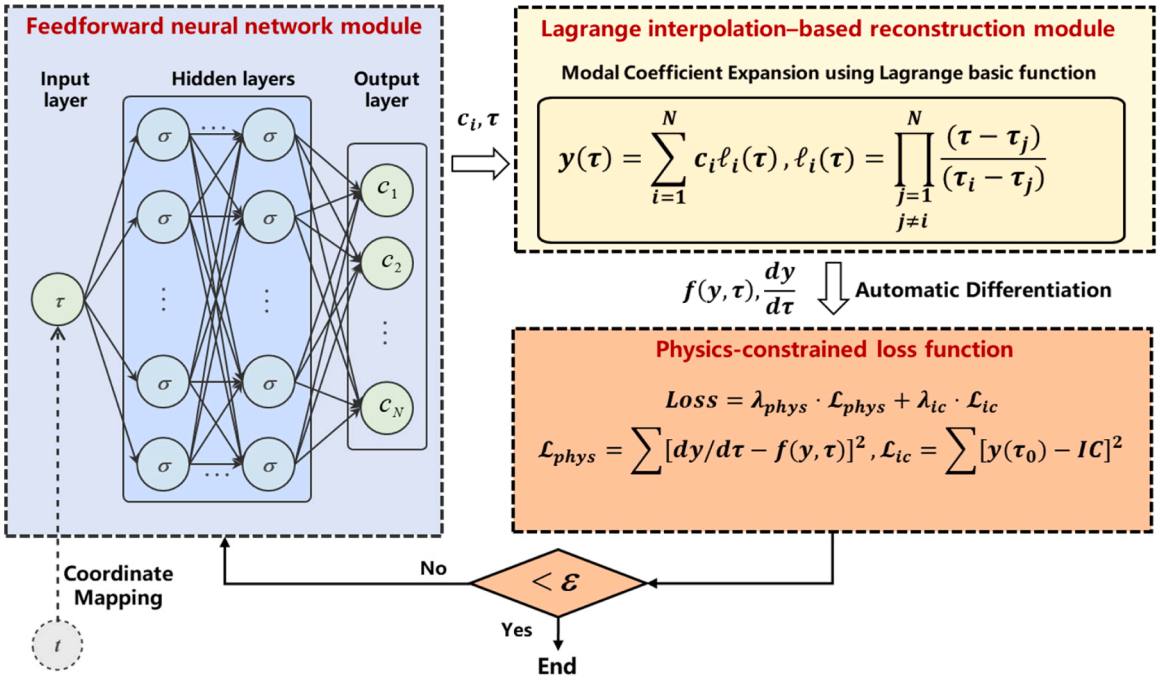


Fig. 1. Schematic diagram of the OCPINN architecture and training workflow.

$$r_{k,d} = \begin{cases} \left.\frac{dy^d}{dt}\right|_{t_k} - f_d(y(t_k), t_k), & \text{for PINN} \\ \left.\frac{dy^d}{d\tau}\cdot\frac{d\tau}{dt}\right|_{\tau_k} - f_d(y(t(\tau_k)), t(\tau_k)), & \text{for OCPINN} \end{cases} \quad (9)$$

where N_d denotes the number of state variables, N_k is the total number of training points, y^d represents the d-th state variable, and τ_k /t_k is the k-th training point. $r_{k,d}$ is the residual of the d-th governing equation at τ_k /t_k. The term $\frac{dy^d}{d\tau}\cdot\frac{d\tau}{dt}|_{\tau_k}$ is obtained by automatic differentiation, while $t(\tau_k)$ denotes the physical time corresponding to τ_k through the inverse of the normalization mapping, also referred to as t_k. Equal weights, $\omega_d = 1/N_d$, are used unless adaptive state-residual weighting is explicitly specified. The adaptive procedure balances the residuals of different state equations, which is introduced in the Supporting Information Section S1. The loss for the initial condition is defined as

$$\mathscr{L}_{\text{ic}} = \frac{1}{N_d}\sum_{d=1}^{N_d} \|y^d(\tau_0) - \hat{y}_0^d\|^2 \quad (10)$$

where $y^d(\tau_0)$ is the predicted value of the d-th variable at the initial time, and $\hat{y}_0^d$ is its corresponding true initial value.

Training is performed using the Adam optimizer (Kingma and Adam, 2017) to minimize $\mathscr{L}_{\text{total}}$. It should be emphasized that no reference solution data are used during training. The parameters of neural networks are determined by minimizing governing-equation residuals together with the prescribed initial-condition treatment. The numerical reference solutions are generated independently and are used only after training to evaluate the accuracy of the PINN and OCPINN predictions.

2.3. Model evaluation metrics

Model accuracy is evaluated on N_e independent staggered evaluation points. These points differ from the N_k residual grid used during training. Reference trajectories are generated using a high-accuracy numerical ODE solver and do not participate in training or checkpoint selection.

To quantify the pointwise prediction accuracy of the model across different time instants, we introduce a time-dependent and state-scaled mean squared error metric $\varepsilon(t_k)$. Because the state variables can have substantially different magnitudes, each state is normalized using a model-independent reference root-mean-square scale s^d. They are defined as

$$\varepsilon(t_k) = \frac{1}{N_d}\sum_{d=1}^{N_d}\left(\frac{y_k^d - \hat{y}_k^d}{s^d}\right)^2, \quad s^d = \frac{1}{N_e}\sum_{k=1}^{N_e}\left(\hat{y}_k^d\right)^2 \quad (11)$$

where y_k^d and $\hat{y}_k^d$ denote the predicted value and true value of the d-th variable at the k-th discrete evaluation point, respectively. $\varepsilon(t_k)$ gives equal importance to all state variables and is used to examine the temporal distribution of the prediction error. The primary scalar accuracy metric is the state-scaled global root-mean-square error, defined as

$$E_{SRMSE} = \sqrt{\frac{1}{N_e}\sum_{k=1}^{N_e}\varepsilon(t_k)} \quad (12)$$

Unless otherwise stated, the principal comparisons and interpolation-node studies were repeated using ten fixed random seeds. The architecture and loss-weight sensitivity analyses used three seeds, as specified in their respective tables. Accuracy metrics are calculated separately for every trained model before statistical aggregation. The results are reported as the mean and sample standard deviation across seeds. Prediction figures show the deterministic reference trajectory together with the mean prediction and its variability across independently trained models. Training time, trainable parameter count, and the frequency of paired wins across identical seeds are reported as complementary computational indicators.

3. Case studies

This section evaluates and compares the performance of the proposed OCPINN method with that of the conventional PINN and VS-PINN across three representative ODE problems (Cases 1-3). Three cases are considered: a nonlinear consecutive-reaction system, the strongly stiff Robertson benchmark, and an industrial methanol-to-propylene (MTP) fixed-bed reactor case. They evaluate OCPINN under nonlinear transient dynamics, pronounced stiffness, and multivariable process coupling. The objective is to compare the predictive accuracy, training behavior, and computational characteristics of OCPINN with PINN and VS-PINN under these different problem settings.

OCPINN, PINN, and VS-PINN are implemented using the PyTorch framework (Paszke et al., 2019) under the Python environment. For fairness, within each case, PINN, VS-PINN, and OCPINN use the same hidden-layer architecture, activation function, learning rate, weight-initialization strategy, number of training epochs, and residual-point grid. The method-specific differences are restricted to their solution representations and, for VS-PINN, the calibrated input-scaling factor. The scaling factor is selected using a held-out physics-loss criterion without access to the reference solution. For the principal comparisons, OCPINN uses $N = 5$ Gauss–Legendre interpolation nodes, while all models are trained using 1000 residual points for 10000 epochs.

The true solutions are obtained using the LSODA solver from the SciPy library (Hindmarsh and Petzold, 2005). To facilitate comparison of model performance across different problem settings, the network architectures and key hyperparameter settings used in Cases 1, 2, and 3 are summarized in Table 1. More details can be found in the code repository. For the Robertson problem, an additional architecture-sensitivity analysis is performed for both PINN and OCPINN, while the main comparison uses the fixed 64-64-64 architecture.

3.1. Case 1: Nonlinear consecutive reaction (Nonlinear ODEs)

In a constant-temperature batch reactor, species A undergoes a first-order reaction to form the intermediate product B, which subsequently undergoes a second-order reaction to yield the final product C. The sequential reaction process can be represented by the following equations:

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C \quad (13)$$

The following ODEs describe the concentration changes of species A and B:

$$\begin{cases} \dfrac{dc_A}{dt} = -k_1 c_A \\ \dfrac{dc_B}{dt} = k_1 c_A - 2k_2 {c_B}^2 \end{cases} \quad (14)$$

Table 1
Network architectures and hyperparameter settings for Cases 1, 2, and 3.

Hyperparameter	Case		
	Case 1	Case 2	Case 3
Hidden layers	3	3	3
Neurons/layer	32-64-32	64-64-64	64-64-64
Activation function	tanh	tanh	tanh
Initial learning rate	10^{-3}	10^{-3}	10^{-3}
Epoch	10,000	10,000	10,000
Number of residual points	1000	1000	1000

where the initial conditions are: $c_A(0) = 1.0$ mol/L, $c_B(0) = 0$ mol/L, $k_1 = 0.1$ min^{-1}, and $k_2 = 0.05$ L/mol/min with the time range $t \in [0, 50]$ min.

Fig. 2 compares the predicted concentrations of species A and B with the true solutions. The curves and shaded regions represent the mean predictions and one standard deviation, respectively, over ten fixed random seeds. All three models reproduce the overall concentration profiles, including the rapid formation and subsequent depletion of intermediate species B. Nevertheless, OCPINN provides the closest and most consistent agreement with the reference solutions, achieving a state-scaled global RMSE of 1.53×10^{-3}, compared with 3.27×10^{-3} for PINN and 2.73×10^{-3} for VS-PINN. Moreover, OCPINN achieves a lower error than both PINN variants for all ten paired random seeds, indicating that its improvement is robust with respect to network initialization.

Fig. 3(a) compares the training-loss histories. OCPINN reaches a lower loss level and exhibits a narrower seed-to-seed variation during the later stages of training. The mean best losses are 8.00×10^{-8}, 3.40×10^{-7}, and 4.64×10^{-7} for OCPINN, PINN, and VS-PINN, respectively. Although VS-PINN reduces the mean prediction error relative to PINN using a calibrated input-scaling factor of 4, its improvement is less consistent across random seeds.

Fig. 3(b) further shows the time-dependent state-scaled MSE. The largest errors for PINN and VS-PINN occur during the initial transient, with peak values of approximately 5.08×10^{-5} and 7.07×10^{-5}, respectively, whereas the maximum mean error of OCPINN is 1.16×10^{-5}. These results indicate that the structured OCPINN representation more accurately captures the rapid transient behavior of the nonlinear consecutive reaction.

To further examine the effect of the proposed representation, Fig. 4 compares the time-varying interpolation coefficients $c_i^{(d)}(t)$ predicted by OCPINN (dashed lines) with the fixed coefficients c_i^{ref} (black dots) obtained from the reference solution (black lines) at the interpolation nodes. As shown in Figs. 4(a) and 4(b), the OCPINN coefficients evolve continuously over time, whereas the conventional fixed-coefficient representation corresponds to a single static interpolation polynomial. The time-varying formulation therefore provides additional flexibility for representing local changes in the solution while retaining the Legendre–Lagrange basis structure.

Figs. 4(c) and 4(d) compare the true concentration slopes with those obtained from the fixed- and time-varying interpolation coefficient reconstructions. The fixed-coefficient interpolation shows noticeable deviations in the local slopes, particularly near the temporal boundaries. The time-varying reconstruction follows the true slopes much more closely. This comparison illustrates how the time-varying coefficient representation improves the description of local transient dynamics while retaining the underlying orthogonal-collocation structure.

3.2. Case 2: Robertson problem (Stiff ODEs)

The Robertson model is a standard test problem in chemical kinetics for systems with stiffness (Hairer and Wanner, 1996). The reaction scheme is as follows:

$$\begin{cases} A \xrightarrow{k_1} B \\ B + B \xrightarrow{k_2} C + B \\ B + C \xrightarrow{k_3} A + C \end{cases} \tag{15}$$

The corresponding ODEs are:

$$\begin{cases} \dfrac{dy_1}{dt} = -k_1 y_1 + k_3 y_2 y_3 \\ \dfrac{dy_2}{dt} = k_1 y_1 - k_2 {y_2}^2 - k_3 y_2 y_3 \\ \dfrac{dy_3}{dt} = k_2 {y_2}^2 \end{cases} \tag{16}$$

The reaction rate constants are $k_1 = 4 \times 10^{-2}$, $k_2 = 3 \times 10^7$, $k_3 = 10^4$, with a range of 9 orders of magnitude, resulting in significant differences in the system. The initial conditions are given by [1, 0, 0], and the time domain is $[0, 10^2]$. To quantify the stiffness of the complete Robertson system, the eigenvalues of the physical Jacobian were evaluated along the numerical reference trajectory. Excluding the zero eigenvalue associated with the conservation relation, the instantaneous stiffness ratio increases from approximately 3.00×10^3 at $t = 10^{-4}$ to 4.70×10^5 at $t = 100$, confirming the pronounced separation between the fast and slow dynamical modes. Representative eigenvalues, stiffness ratios, and reference-solver diagnostics are reported in Supporting Information Section S2.

Because the Robertson dynamics span multiple time scales and the intermediate species remains several orders of magnitude smaller than the dominant states, a shifted logarithmic time coordinate is used to provide denser residual coverage of the rapid initial transient while retaining the complete physical time domain. A constraint-aware output transformation is also introduced so that the initial condition, nonnegativity, and conservation relation are satisfied by construction. The residuals of all three original governing equations are evaluated in physical time using automatic differentiation and the chain rule. The same coordinate transformation and constraint treatment are applied to PINN, VS-PINN, and OCPINN to ensure a consistent comparison;

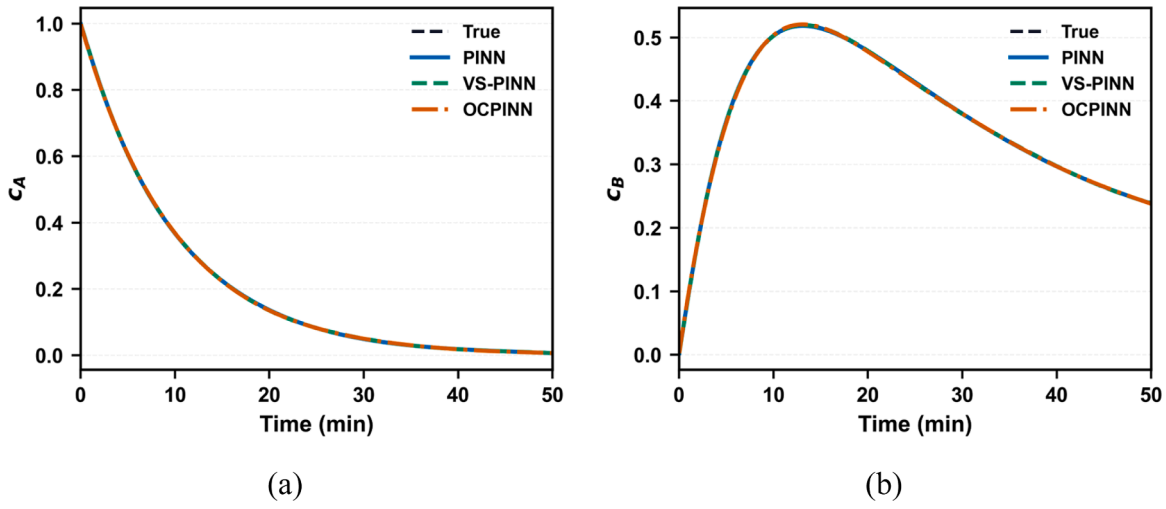


Fig. 2. Case 1: Comparison of predicted solutions by PINN, VS-PINN and OCPINN in the batch reactor system with true solutions for (a) c_A and (b) c_B.

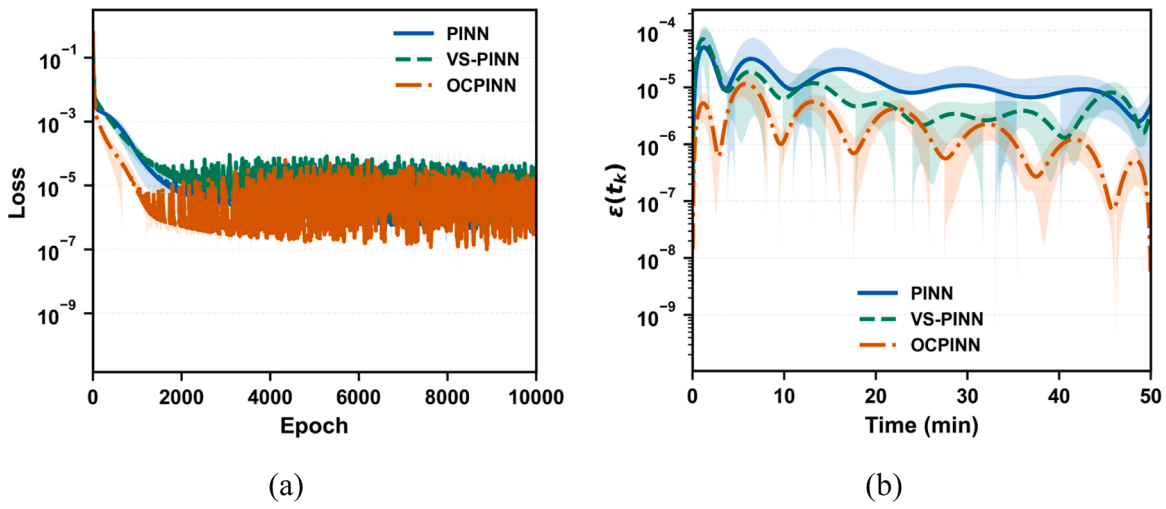


Fig. 3. Case 1: (a) Training loss convergence curves for PINN, VS-PINN, and OCPINN; (b) Comparison of the time-dependent error curves $\varepsilon(t_k)$ obtained by PINN, VS-PINN, and OCPINN.

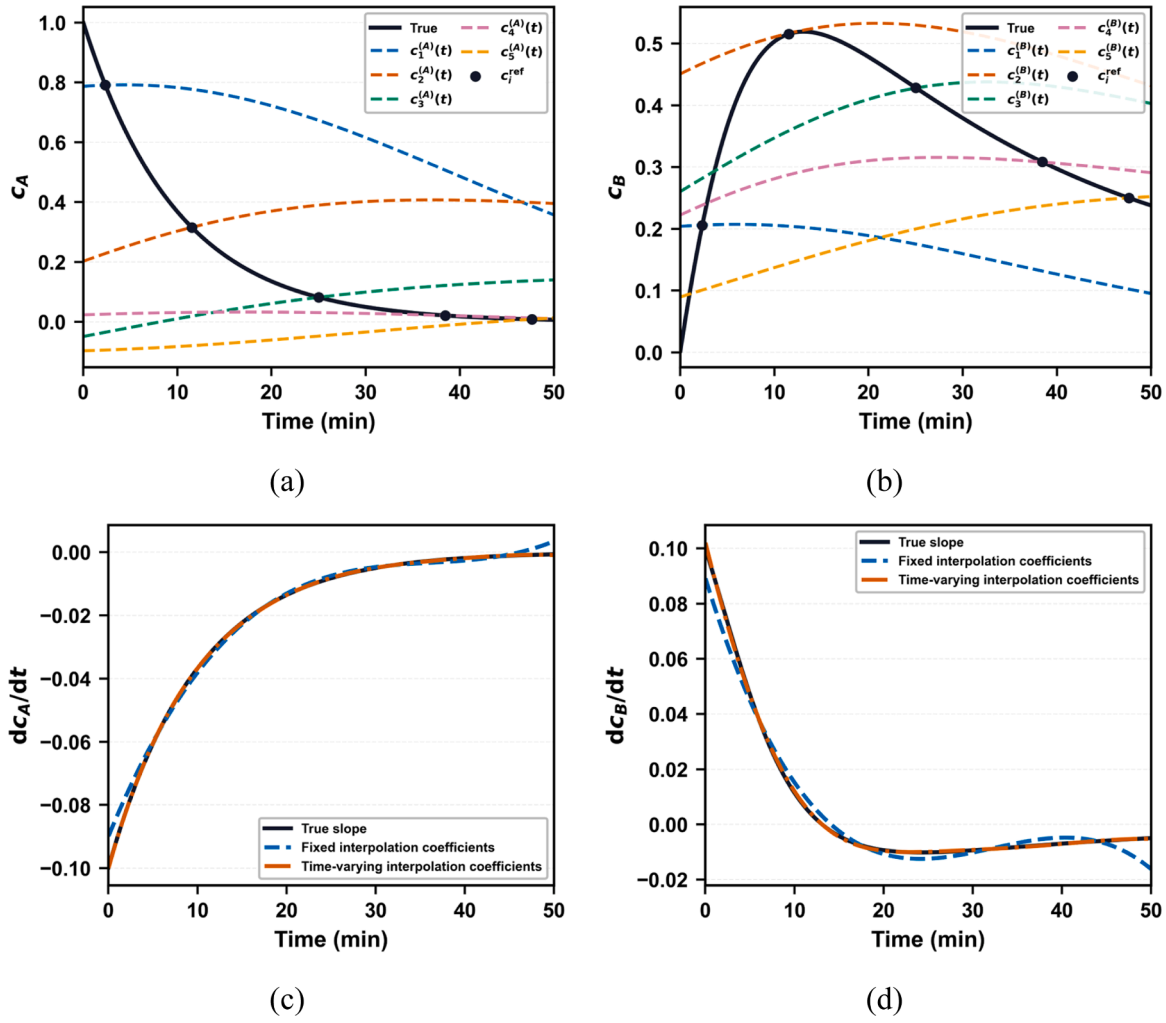


Fig. 4. Comparison of interpolation coefficients in OCPINN for Case 1. (a, b) Time evolution of interpolation coefficients compared with the true solution; (c, d) Instantaneous slopes of fixed and time-varying coefficient reconstructions.

implementation details are given in Supporting Information Section S2.

Fig. 5 compares the ten-seed mean predictions with the numerical reference solution. All three methods reproduce the multi-timescale Robertson dynamics, including the slow depletion of y_1, the rapid formation of the intermediate species y_2, and the accumulation of y_3. The principal discrepancy occurs during the fast transient of y_2, where the

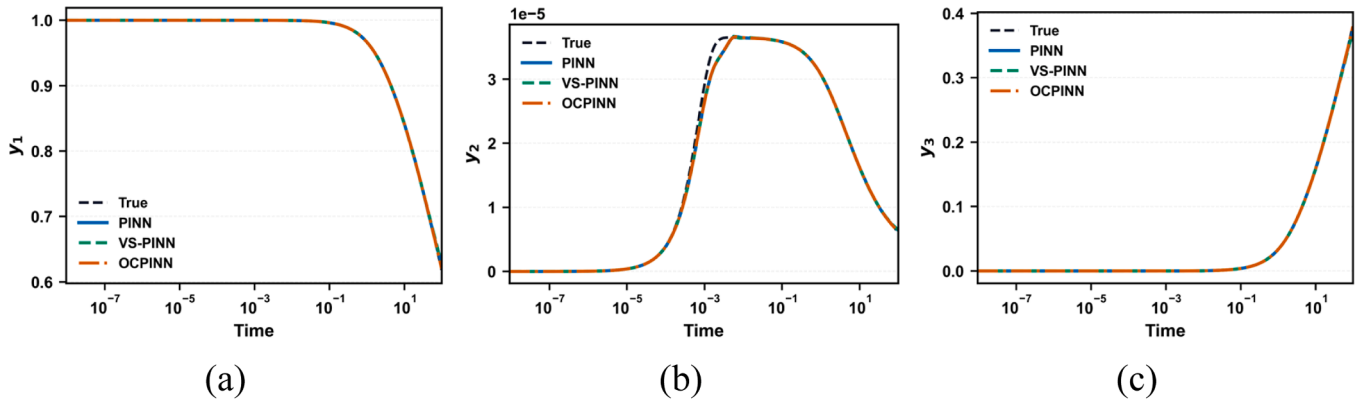


Fig. 5. Case 2: Comparison of predicted solutions by PINN, VS-PINN, and OCPINN in the Robertson problem for (a) y_1, (b) y_2 and (c) y_3 with true solutions.

predicted transition is slightly delayed relative to the reference solution. OCPINN provides a clearer improvement for the slowly evolving components: the mean relative L_2 errors of OCPINN for y_1 and y_3 are 3.27×10^{-4} and 3.08×10^{-3}, respectively, compared with 8.23×10^{-4} and 7.75×10^{-3} for PINN.

Fig. 6 further compares the training-loss and time-dependent error behavior. As shown in Fig. 6(a), all three methods converge to similar final loss levels of approximately 2.04×10^{-5}, although VS-PINN and OCPINN decrease somewhat faster during the early training stage. Fig. 6 (b) shows that the dominant prediction error is concentrated around the rapid transient of the intermediate species, while substantially smaller errors are observed over most of the remaining time domain. Over the complete evaluation domain, OCPINN achieves the lowest mean state-scaled global RMSE of 2.062×10^{-2}, compared with 2.097×10^{-2} for PINN and 2.133×10^{-2} for VS-PINN. Thus, the global improvement is modest for this strongly stiff benchmark, but the component-wise results indicate a more noticeable advantage for the slowly evolving states y_1 and y_3.

To examine whether this comparison depends strongly on network capacity, PINN and OCPINN are additionally evaluated using five representative architectures under otherwise identical training settings. The complete architecture-sensitivity results are summarized in Table 2, with the corresponding PINN results provided in Supporting Information Section S2. The 128-128-128 architecture gives a slightly lower mean error than the 64-64-64 network, but at approximately four times the parameter count. The 64-64-64 architecture is therefore retained as a compact reference configuration rather than as a universally optimal choice. Across the five architectures, OCPINN achieves a lower mean error than PINN in four cases, indicating that its comparative performance is not restricted to a single network size.

Table 2
Architecture sensitivity of OCPINN for the Robertson problem.

Architectures	E_{SRMSE} [#]	Parameters	Training Time (s) [#]
64-64	$(2.125 \pm 0.043) \times 10^{-2}$	4938	410.6 ± 12.8
32-32-32	$(2.400 \pm 0.386) \times 10^{-2}$	2506	431.9 ± 21.0
64-64-64	$(2.050 \pm 0.006) \times 10^{-2}$	9098	431.9 ± 30.4
128-128-128	$(2.045 \pm 0.003) \times 10^{-2}$	34,570	420.6 ± 3.0
64-64-64-64	$(2.062 \pm 0.009) \times 10^{-2}$	13,258	443.0 ± 10.0

[#] Values are reported as the mean ± standard deviation over three fixed random seeds. Training time represents the wall-clock duration of the 10,000-epoch optimization loop.

3.3. Case 3: An industrial MTP fixed-bed reactor case

To further demonstrate the applicability of OCPINN in process-relevant reaction systems, the proposed OCPINN is applied to a methanol-to-propylene (MTP) fixed-bed reactor model (Zhang et al., 2025). Specifically, a single-stage model from the original reactor optimization framework is formulated as an ODE initial-value problem along the reactor bed coordinate, where a coupled reaction network and energy balance govern the evolution of the molar flow rates of the lumped species and the reactor temperature. More details about the

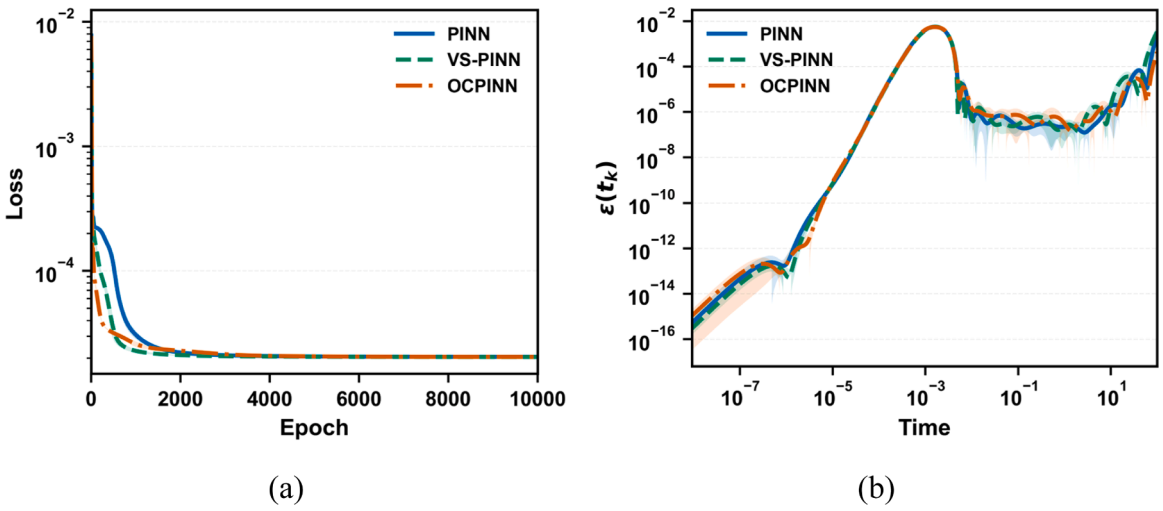


Fig. 6. Case 2: (a) Training loss convergence curves of PINN, VS-PINN, and OCPINN; (b) Comparison of the time-dependent error curves $\varepsilon(t_k)$ obtained by PINN, VS-PINN and OCPINN.

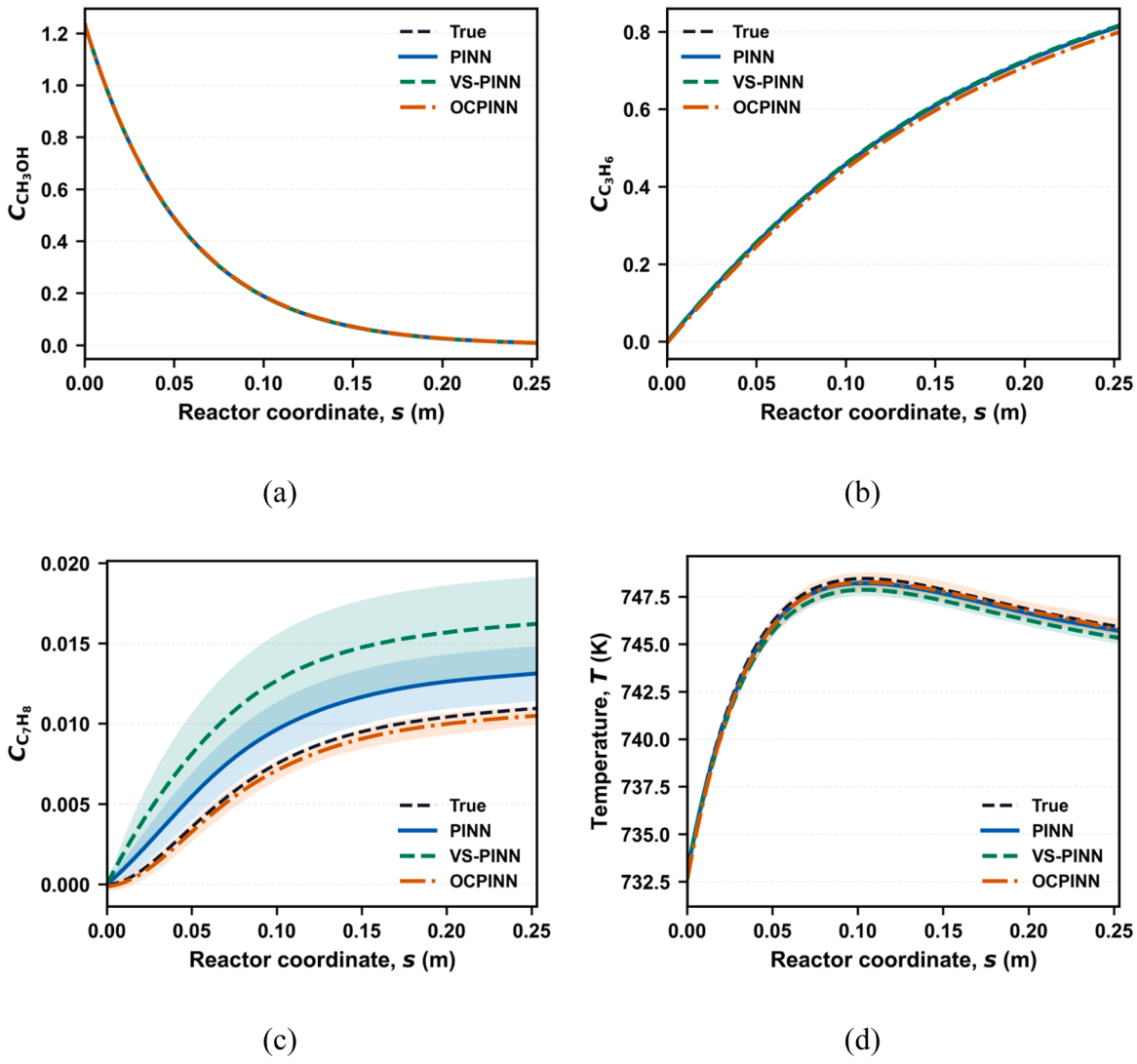


Fig. 7. Comparison of predicted key-variable trajectories along the reactor coordinate (a) CH_3OH; (b) C_3H_6; (c) C_7H_8; (d) temperature.

model can be found from the reference (Zhang et al., 2025). The component balances for the selected species and the energy balance are written as:

$$\begin{cases} \dfrac{dC_{CH_3OH}}{ds} = -\kappa(r_1 + r_2 + r_3 + r_4 + r_5 + r_6 + r_7 + r_8 + r_9 + r_{10}) \\ \dfrac{dC_{C_2H_4}}{ds} = \kappa\left(\dfrac{1}{2}r_1 + r_{14}\right) \\ \dfrac{dC_{C_3H_6}}{ds} = \kappa\left(\dfrac{1}{3}r_2 - r_7 + r_{11} + 2r_{12} + 2r_{13} + r_{15} - r_{17}\right) \\ \dfrac{dC_{C_4H_8}}{ds} = \kappa\left(\dfrac{1}{4}r_3 + r_7 - r_8 - 2r_{11} + r_{12} + r_{14} + r_{15}\right) \\ \dfrac{dC_{C_5H_{10}}}{ds} = \kappa\left(\dfrac{1}{5}r_4 + r_8 - r_9 + r_{11} - 2r_{12}\right) \\ \dfrac{dC_{C_6H_{12}}}{ds} = \kappa\left(\dfrac{1}{6}r_5 + r_9 - r_{10} - r_{13} - r_{14}\right) \\ \dfrac{dC_{C_7H_{14}}}{ds} = \kappa\left(\dfrac{1}{7}r_6 + r_{10} - r_{15} - r_{16}\right) \\ \dfrac{dC_{C_7H_8}}{ds} = \kappa r_{16} \\ \dfrac{dC_{C_3H_8}}{ds} = \kappa r_{17} \\ \dfrac{dC_{H_2O}}{ds} = \kappa(r_1 + r_2 + r_3 + r_4 + r_5 + r_6 + r_7 + r_8 + r_9 + r_{10}) \\ \dfrac{dT}{ds} = -\rho\left(\dfrac{\pi d_t^{\,2}}{4}\right)\dfrac{\sum_{n=1}^{17} r_n \Delta H_n}{\overline{C_p}\cdot \dot{m}} \end{cases} \tag{17}$$

where C_i (mol/(m^3)) denotes the concentration of lumped component i, T (K) denotes temperature, s denotes the reactor coordinate, and r_n (mol/(kg$_{cat}$ • h)) denotes the kinetic rate expressions of reaction n. ΔH_n (kJ/mol) represents the effective heat-release contribution associated with reaction n, $\overline{C_p}$ (kJ/(kg • K)) denotes the heat capacity, and $\dot{m}$ (kg/h) represents the mass flow rate. κ is a defined parameter, which is equal to $\frac{1}{Q_V}\left(\frac{\pi d_t^{\,2}}{4}\right)\rho$, where Q_V (m^3/h) is gas volume flowrate, d_t (m) is the reactor diameter, and ρ (kg/m^3) is the reactor bed density. The complete kinetic scheme, model parameters, initial conditions, and coordinate specifications are provided in the Supporting Information Section S3.

Fig. 7 compares the predicted axial profiles of representative variables, including CH_3OH, C_3H_6, C_7H_8, and temperature, with the numerical reference solution. The remaining state trajectories are reported in the Supporting Information Section S3. All three methods reproduce the dominant reactor trends, although their accuracy differs across individual state variables; OCPINN provides the lowest overall prediction error. Under the common $N = 5$ setting, OCPINN achieves a ten-seed state-scaled global RMSE of $(4.208 \pm 1.515) \times 10^{-2}$, compared with $(1.339 \pm 1.045) \times 10^{-1}$ for PINN, and $(3.206 \pm 1.799) \times 10^{-1}$ for VS-PINN. OCPINN outperforms PINN in seven of the ten paired runs and VS-PINN in all ten runs.

The improvement is most pronounced for the low-magnitude intermediate components. For C_7H_{14} and C_7H_8, the mean relative errors decrease from 0.367 to 0.248 for PINN to 0.0864 and 0.0659 for OCPINN, respectively. This indicates that the structured reconstruction is particularly beneficial for states that are more difficult to resolve accurately within the multivariable reactor system. Training-loss histories, time-dependent error distributions, and the complete set of predicted trajectories are provided in Supporting Information Section S3.

The principal MTP comparison uses fixed equal state-residual weights, because an adaptive-weighting ablation applied consistently to all methods does not improve the state-scaled global accuracy. The complete results are provided in the Supporting Information Section S3.

4. Further analysis and discussion

4.1. Influence of the number of interpolation nodes

Table 3 summarizes the state-scaled global RMSE, trainable parameter count, and training time of OCPINN with $N = 3 \sim 7$ Gauss–Legendre interpolation nodes, while Fig. 8 illustrates the corresponding accuracy trends. The influence of N is strongly problem dependent. For Case 1, increasing the number of interpolation nodes consistently improves prediction accuracy: the mean global RMSE decreases from $(2.079 \pm 0.851) \times 10^{-3}$ at $N = 3$ to $(0.518 \pm 0.190) \times 10^{-3}$ at $N = 7$. All investigated OCPINN configurations outperform the PINN and VS-PINN baselines. At the common reference setting of $N = 5$, the mean error is

Table 3
State-scaled global RMSE, parameter count, and training time for OCPINN with different numbers of interpolation nodes, with PINN, parameter-matched PINN, and VS-PINN as reference baselines.

Cases	Methods	E_{SRMSE} [#]	Parameters	Training Time (s) [#]
Case 1	OCPINN, $N = 3$	$(2.079 \pm 0.851) \times 10^{-3}$	4454	266.9 ± 14.2
	OCPINN, $N = 4$	$(1.753 \pm 0.557) \times 10^{-3}$	4520	292.1 ± 20.3
	OCPINN, $N = 5$	$(1.526 \pm 0.277) \times 10^{-3}$	4586	350.4 ± 16.0
	OCPINN, $N = 6$	$(0.907 \pm 0.354) \times 10^{-3}$	4652	382.9 ± 14.4
	OCPINN, $N = 7$	$(0.518 \pm 0.190) \times 10^{-3}$	4718	455.9 ± 14.2
	PINN	$(3.269 \pm 1.573) \times 10^{-3}$	4322	178.7 ± 18.0
	PINN (Matched)	$(4.216 \pm 1.355) \times 10^{-3}$	4590	177.8 ± 10.6
	VS-PINN	$(2.724 \pm 0.796) \times 10^{-3}$	4322	181.7 ± 13.9
Case 2	OCPINN, $N = 3$	$(2.092 \pm 0.047) \times 10^{-2}$	8838	299.5 ± 14.3
	OCPINN, $N = 4$	$(2.065 \pm 0.012) \times 10^{-2}$	8968	348.0 ± 12.6
	OCPINN, $N = 5$	$(2.062 \pm 0.015) \times 10^{-2}$	9098	401.8 ± 9.0
	OCPINN, $N = 6$	$(2.069 \pm 0.015) \times 10^{-2}$	9228	479.3 ± 8.0
	OCPINN, $N = 7$	$(2.055 \pm 0.006) \times 10^{-2}$	9358	580.8 ± 12.7
	PINN	$(2.097 \pm 0.024) \times 10^{-2}$	8578	260.1 ± 10.6
	PINN (Matched)	$(2.087 \pm 0.016) \times 10^{-2}$	9114	259.7 ± 11.9
	VS-PINN	$(2.133 \pm 0.093) \times 10^{-2}$	8578	263.8 ± 7.4
Case 3	OCPINN, $N = 3$	$(2.021 \pm 0.981) \times 10^{-2}$	10,593	625.1 ± 11.5
	OCPINN, $N = 4$	$(4.277 \pm 2.164) \times 10^{-2}$	11,308	751.2 ± 9.2
	OCPINN, $N = 5$	$(4.208 \pm 1.515) \times 10^{-2}$	12,023	921.8 ± 16.2
	OCPINN, $N = 6$	$(15.771 \pm 7.782) \times 10^{-2}$	12,738	1137.5 ± 16.5
	OCPINN, $N = 7$	$(29.053 \pm 10.610) \times 10^{-2}$	13,453	1413.3 ± 25.4
	PINN	$(13.386 \pm 10.445) \times 10^{-2}$	9163	432.1 ± 11.0
	PINN (Matched)	$(1.647 \pm 0.695) \times 10^{-1}$	12,051	435.8 ± 14.8
	VS-PINN	$(32.063 \pm 17.988) \times 10^{-2}$	9163	442.1 ± 14.4

[#] Values are reported as the mean ± standard deviation over ten fixed random seeds. Training time represents the wall-clock duration of the 10,000-epoch optimization loop.

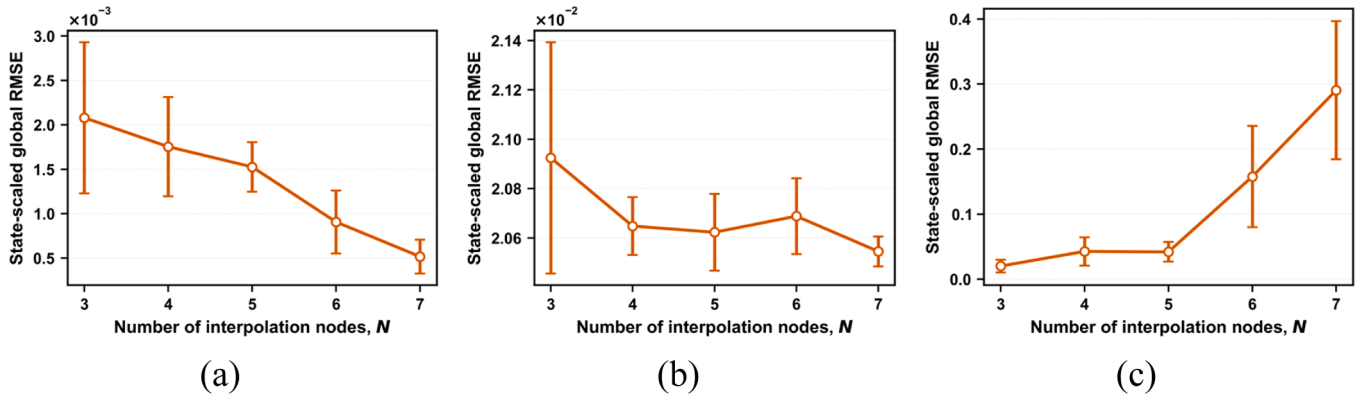


Fig. 8. State-scaled global RMSE of OCPINN as the number of interpolation nodes N increases for (a) Case 1, (b) Case 2, and (c) Case 3.

reduced by approximately 53.3% and 44.0% relative to PINN and VS-PINN, respectively.

The Robertson problem shows a markedly different dependence on N. As shown in Fig. 8(b), the mean errors remain within a narrow range of 2.055×10^{-2} to 2.092×10^{-2}, with only small and non-monotonic variations among the investigated configurations. For the MTP reactor, the dependence is stronger but follows the opposite trend to Case 1. The lowest mean error, $(2.021 \pm 0.981) \times 10^{-2}$, is obtained at $N = 3$; accuracy deteriorates substantially at $N = 6$ and $N = 7$. These results indicate that increasing the number of interpolation nodes does not necessarily translate into improved trained-model accuracy.

Increasing N also raises the output dimension, parameter count, and cost of residual evaluation. Across $N = 3$ to $N = 7$, the mean training time increases from 266.9 to 455.9 s for Case 1, from 299.5 to 580.8 s for Case 2, and from 625.1 to 1413.3 s for Case 3. A larger reconstruction space therefore provides greater approximation capacity at the expense of a larger optimization problem and higher automatic-differentiation cost. Accordingly, N should be regarded as a structural hyperparameter that balances representation capacity, optimization difficulty, and computational cost, rather than as a conventional discretization order for which increasing N guarantees monotonic error reduction.

4.2. Model capacity and representation analysis

Compared with a conventional PINN that directly maps the independent coordinate to the state variables, OCPINN predicts multiple interpolation coefficients for each state, resulting in a modest increase in the size of the output mapping.

To determine whether the observed accuracy improvement is simply a consequence of this increased model capacity, a parameter-matched PINN is constructed by increasing the width of its final hidden layer while keeping the network depth, activation function, learning rate, residual points, and other training settings unchanged. The resulting model contains approximately the same number of trainable parameters as, or slightly more parameters than, the $(N = 5)$ OCPINN configuration.

As summarized in Table 3, increasing the parameter count of PINN alone does not produce a systematic improvement in predictive accuracy. The parameter-matched PINN performs worse than the original PINN in Cases 1 and 3 and provides only a marginal improvement in the Robertson case. In contrast, OCPINN achieves a lower state-scaled global RMSE than the parameter-matched PINN in 29 of the 30 paired random-seed runs. The improvement is substantial for the nonlinear consecutive-reaction and MTP reactor cases and smaller, but consistent, for the stiff Robertson problem. These results indicate that the advantage of OCPINN cannot be attributed solely to its increased parameter count, but is associated with the time-varying coefficient representation and Legendre–Lagrange reconstruction.

The structured reconstruction introduces additional computational cost. Despite having parameter counts comparable to the matched PINN, the $(N = 5)$ OCPINN requires approximately 1.5–2.1 times longer training because residual evaluation involves both Lagrange reconstruction and automatic differentiation through multiple coefficient functions. Classical LSODA remains substantially more efficient for one-off forward integration; accordingly, OCPINN is regarded here as a continuously differentiable solution representation rather than a replacement for established ODE integrators. Detailed runtime and inference benchmarks are provided in Supporting Information Section S5.

4.3. Approximation capacity of the OCPINN reconstruction

The classical Legendre–Lagrange interpolant is contained within the OCPINN hypothesis space as a special case by setting the coefficient functions equal to the corresponding constant nodal values. Consequently, for a reference solution that is analytic in a neighborhood of the normalized domain, the best achievable approximation within the OCPINN hypothesis space satisfies

$$\inf_{\boldsymbol{\theta}} \|y_{\mathrm{ref}} - \widehat{y}_N(\cdot;\boldsymbol{\theta})\|_\infty \le C_r r^{-N}, \quad 1 < r < \rho \tag{19}$$

This result establishes the spectral approximation potential of the reconstruction space rather than exponential convergence of the trained model. The actual prediction error additionally depends on how accurately the neural network learns the coefficient functions. As derived in Supporting Information Section S4,

$$\| y_{\mathrm{ref}} - \widehat{y}_N\|_\infty \le C_r r^{-N} + \Lambda_N \max_i \|y_{\mathrm{ref}}(\tau_i) - c_i(\cdot;\boldsymbol{\theta})\|_\infty \tag{20}$$

where Λ_N is the Lebesgue constant associated with the interpolation nodes. The first term represents the approximation error associated with the Legendre–Lagrange basis, whereas the second reflects coefficient-learning and optimization errors. Thus, increasing N enlarges the approximation capacity of the reconstruction but does not guarantee a monotonic or exponential decrease in the trained-model error, consistent with the problem-dependent behavior observed in Table 3 and Fig. 8. Accordingly, N is treated as a structural hyperparameter rather than a conventional discretization order.

For the Robertson problem, the reconstructed latent variables are subsequently passed through a nonlinear constraint-aware transformation. The above approximation-capacity result therefore applies directly to the Legendre–Lagrange reconstruction layer and should not be interpreted as a direct error bound for the final constrained physical states.

4.4. Limitations

Each trained model represents the solution for specified governing equations, parameters, and initial conditions within a prescribed

domain. Neither out-of-domain extrapolation nor generalization across operating conditions is evaluated. The Legendre–Lagrange reconstruction is defined on the normalized interval [−1,1], and the associated approximation properties are therefore not guaranteed for extrapolation beyond this interval. Although the differentiable representation may facilitate sensitivity analysis and optimization workflows, these application-level benefits are not demonstrated here.

The current formulation is limited to forward initial-value problems governed by ODE systems. Its applicability to boundary-value problems, PDEs, and other distributed-parameter systems has not been investigated. Such extensions would require additional treatment of boundary constraints, multidimensional basis construction, residual-point allocation, and computational scaling.

5. Conclusion

In this study, we develop OCPINN, a physics-informed neural representation that incorporates a Legendre–Lagrange reconstruction layer with neural-network-predicted time-varying interpolation coefficients. Unlike a conventional PINN, OCPINN reconstructs the continuous solution through a structured basis representation while retaining residual-based training and automatic differentiation.

The method is evaluated on a nonlinear consecutive-reaction system, the complete stiff Robertson system, and an industrial methanol-to-propylene fixed-bed reactor. Across the three cases, OCPINN achieves the lowest mean state-scaled global RMSE among the conventional PINN and VS-PINN baselines. The improvement is pronounced for the nonlinear reaction and MTP reactor cases, whereas the advantage for the Robertson problem is modest at the global level. Parameter-matched comparisons further show that the observed improvement cannot be attributed solely to the increased number of trainable parameters introduced by the reconstruction layer.

The influence of the number of interpolation nodes was problem dependent, reflecting a tradeoff between representation capacity and computational cost. OCPINN is best regarded as a differentiable structured solution representation rather than a replacement for established ODE solvers in one-off forward simulations. The present study is limited to forward initial-value problems within prescribed domains; extensions to inverse problems, parameterized surrogate modeling, boundary-value problems, and distributed-parameter systems remain subjects for future investigation.

CRediT authorship contribution statement

Pengfei Pu: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Xiaodong Hong:** Visualization, Validation, Supervision, Funding acquisition, Conceptualization. **Zuwei Liao:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Jingdai Wang:** Project administration, Funding acquisition. **Yongrong Yang:** Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.compchemeng.2026.109938.

Data availability

The data and code are provided in the code repository. (https://github.com/ZJUChEPSE/OCPINN-public.git).

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